

DPP - 5

- Q.1 Indicate the correct order sequence in respect of the following :
- I. If the probability that a computer will fail during the first hour of operation is 0.01, then if we turn on 100 computers, exactly one will fail in the first hour of operation.
 - II. A man has ten keys only one of which fits the lock. He tries them in a door one by one discarding the one he has tried. The probability that fifth key fits the lock is $1/10$.
 - III. Given the events A and B in a sample space. If $P(A) = 1$, then A and B are independent.
 - IV. When a fair six sided die is tossed on a table top, the bottom face can not be seen. The probability that the product of the numbers on the five faces that can be seen is divisible by 6 is one.
- (A) FTFT (B) FTTT (C) TFTF (D) TFFF
- Q.2 If a, b and c are three numbers (not necessarily different) chosen randomly and with replacement from the set $\{1, 2, 3, 4, 5\}$, the probability that $(ab + c)$ is even, is
- (A) $\frac{35}{125}$ (B) $\frac{59}{125}$ (C) $\frac{64}{125}$ (D) $\frac{75}{125}$
- Q.3 A examination consists of 8 questions in each of which one of the 5 alternatives is the correct one. On the assumption that a candidate who has done no preparatory work chooses for each question any one of the five alternatives with equal probability, the probability that he gets more than one correct answer is equal to
- (A) $(0.8)^8$ (B) $3(0.8)^8$ (C) $1 - (0.8)^8$ (D) $1 - 3(0.8)^8$
- Q.4 An ant is situated at the vertex A of the triangle ABC. Every movement of the ant consists of moving to one of other two adjacent vertices from the vertex where it is situated. The probability of going to any of the other two adjacent vertices of the triangle is equal. The probability that at the end of the fourth movement the ant will be back to the vertex A, is :
- (A) $\frac{4}{16}$ (B) $\frac{6}{16}$ (C) $\frac{7}{16}$ (D) $\frac{8}{16}$
- Q.5 A key to room number C_3 is dropped into a jar with five other keys, and the jar is thoroughly mixed. If keys are randomly drawn from the jar without replacement until the key to room C_3 is chosen, then what are the odds in favour that the key to room C_3 will be obtained on the 2nd try?
- (A) 1:4 (B) 1:5 (C) 1:6 (D) 5:6
- Q.6 Lot A consists of 3G and 2D articles. Lot B consists of 4G and 1D article. A new lot C is formed by taking 3 articles from A and 2 from B. The probability that an article chosen at random from C is defective, is
- (A) $1/3$ (B) $2/5$ (C) $8/25$ (D) none
- Q.7 Mr. A and Mr. B each have a bag that contains one ball of each of the colours blue, green, orange, red and violet. 'A' randomly selects one ball from his bag and puts it into B's bag. 'B' then randomly selects one ball from his bag and puts it into A's bag. The probability that after this process the contents of the two bags are the same, is
- (A) $1/6$ (B) $1/5$ (C) $1/3$ (D) $1/2$
- Q.8 On a Saturday night 20% of all drivers in U.S.A. are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a Saturday night smashed into a tree, the probability that the driver was under the influence of alcohol, is
- (A) $3/7$ (B) $4/7$ (C) $5/7$ (D) $6/7$
- Q.9 A box has four dice in it. Three of them are fair dice but the fourth one has the number five on all of its faces. A die is chosen at random from the box and is rolled three times and shows up the face five on all the three occasions. The chance that the die chosen was a rigged die, is
- (A) $\frac{216}{217}$ (B) $\frac{215}{219}$ (C) $\frac{216}{219}$ (D) none

- Q.10 A real estate man has eight master keys to open several new houses. Only one master key will open a given house. If 40% of these homes are usually left unlocked, the probability that the real estate man can get into a specific home if he selects three master keys at random, is
 (A) $1/2$ (B) $5/8$ (C) $2/3$ (D) $3/4$
- Q.11 A purse contains 100 coins of unknown value, a coin drawn at random is found to be a rupee, The chance that it is the only rupee in the purse, is (Assume all numbers of rupee coins in the purse to be equally likely.)
 (A) $\frac{1}{5050}$ (B) $\frac{2}{5151}$ (C) $\frac{1}{4950}$ (D) $\frac{2}{4950}$
- Q.12 A purse contains 2 six sided dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is
 (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{24}$
- Q.13 An instrument consists of two units. Each unit must function for the instrument to operate. The reliability of the first unit is 0.9 & that of the second unit is 0.8. The instrument is tested & fails. The probability that "only the first unit failed & the second unit is sound" is :
 (A) $1/7$ (B) $2/7$ (C) $3/7$ (D) $4/7$
- Q.14 A box contains 10 tickets numbered from 1 to 10. Two tickets are drawn one by one without replacement. The probability that the "difference between the first drawn ticket number and the second is not less than 4" is
 (A) $\frac{7}{30}$ (B) $\frac{14}{30}$ (C) $\frac{11}{30}$ (D) $\frac{10}{30}$

Paragraph for question nos. 15 to 17

A JEE aspirant estimates that she will be successful with an 80 percent chance if she studies 10 hours per day, with a 60 percent chance if she studies 7 hours per day and with a 40 percent chance if she studies 4 hours per day. She further believes that she will study 10 hours, 7 hours and 4 hours per day with probabilities 0.1, 0.2 and 0.7, respectively

- Q.15 The chance she will be successful, is
 (A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58
- Q.16 Given that she is successful, the chance she studied for 4 hours, is
 (A) $\frac{6}{12}$ (B) $\frac{7}{12}$ (C) $\frac{8}{12}$ (D) $\frac{9}{12}$
- Q.17 Given that she does not achieve success, the chance she studied for 4 hour, is
 (A) $\frac{18}{26}$ (B) $\frac{19}{26}$ (C) $\frac{20}{26}$ (D) $\frac{21}{26}$

[SUBJECTIVE]

- Q.18 A and B each throw simultaneously a pair of dice. Find the probability that they obtain the same score.
- Q.19 If mn coins have been distributed into m purses, n into each find
 (1) the chance that two specified coins will be found in the same purse, and
 (2) what the chance becomes when r purses have been examined and found not to contain either of the specified coins.
- Q.20 A, B are two inaccurate arithmeticians whose chance of solving a given question correctly are $(1/8)$ and $(1/12)$ respectively. They solve a problem and obtained the same result. If it is 1000 to 1 against their making the same mistake, find the chance that the result is correct.